# 📋 Document 1: Product Requirements Document (PRD)

**Product Name:** ClinikFlow (India Specialist Edition)
**Target Market:** Standalone Single-Doctor Specialist Outpatient Clinics in India
**Document Version:** 1.0.0
**Status:** Approved for Implementation
**Target Form Factors:** Mobile-First PWA (Doctor Mobile/Tablet, Receptionist Mobile/Desktop, Waiting Room Smart TV)

## 1. Executive Summary & Vision

In India, over 70% of outpatient healthcare is delivered through standalone single-doctor clinics. A specialist doctor (Pediatrician, Dermatologist, Orthopedic, Gynecologist, Cardiologist, or Diabetologist) works in high-pressure 3-to-4-hour OPD shifts, seeing 30 to 60 patients.

Existing clinic software fails in India because:

1. **Desktop-heavy:** Demands a full desktop PC inside the doctor's cabin when doctors prefer using an iPad or smartphone.
2. **Slow intake:** Western appointment software takes 2–3 minutes to register a patient, causing front-desk queues to spill onto the street.
3. **Ignores Indian OPD dynamics:** Walk-ins (70%+), quick "report checks" ("*Doctor sahab ko bas 2 min report dikhani hai*"), cash/UPI fee collection at the counter, and WhatsApp-first communication.

**ClinikFlow** is an ultra-fast, mobile-first clinic queue management and lightweight electronic medical record (EMR Lite) platform designed specifically for the single-doctor Indian specialist clinic.

## 2. Key Personas & Ground Realities

### Persona 1: Dr. Rajesh Iyer (The Solo Specialist Doctor)

- **Demographics:** 43-year-old Pediatrician / Diabetologist in a Tier-1/2 Indian city.

- **Clinic Setup:** Private clinic with 1 consultation cabin, 1 waiting area (15 seats), 1 compounder/receptionist.
- **Pain Points:**
    - Hates typing long clinical notes on a keyboard while talking to anxious parents/patients.
    - Loses track of who is waiting outside and gets interrupted by patients walking in unannounced.
    - Wants past medical records and vaccine/blood sugar trends at a glance on his tablet/phone.
- **Success Metric:** Complete a consultation, review past history, and generate an Indian-standard prescription in under 60 seconds on a mobile device.

### Persona 2: Ramesh (The Front-Desk Compounder / Receptionist)

- **Demographics:** 28-year-old clinic assistant managing patient queue, cash/UPI collection, vitals, and crowd control.
- **Pain Points:**
    - Faces constant badgering: "*Hamara number kab aayega?*" (When is our turn?).
    - Manages both cash and UPI payments without an easy daily reconciliation ledger.
    - Cannot spend more than 15 seconds entering patient details during peak evening rush (6:00 PM – 9:00 PM).
- **Success Metric:** Token generated, payment recorded, and WhatsApp token link sent in less than 12 seconds per patient.

### Persona 3: Smt. Sunita & Patient Family (The Indian Patient)

- **Pain Points:**
    - Waiting 90 minutes in a crowded, noisy waiting room with no estimate of when the doctor will call them.
    - Does not want to download another 50MB mobile app or remember a login password.
- **Success Metric:** Receives a WhatsApp message with token number and a live web link ( `clinic.in/q/24` ) showing live position and estimated wait time.

## 3. Product Principles & Tenets

1. **Zero-App Friction for Patients:** No app store downloads, no passwords. Everything runs via WhatsApp and instant web links.
2. **Speed Over Complexity:** The compounder must issue a token in under 12 seconds; the doctor must write an Rx in under 60 seconds.
3. **Touch-First for Doctors:** Every clinical action (vitals, diagnoses, prescription dosages) is achievable via tap chips and quick-selectors on a tablet or phone.
4. **Resilient to Flaky Internet:** The clinic runs locally even during broadband downtime, syncing automatically when connectivity resumes.
5. **Culturally Grounded in Indian Clinic Behavior:** Native support for walk-ins, "report review" re-entries, dynamic UPI QR codes, and vernacular audio announcements.

## 4. Epic & Feature Requirements

### Epic 1: High-Speed Front-Desk Token Generation (< 12 Seconds)

- **FR-1.1 Mobile Number Lookup & UHID Reuse:**
  - Single input field for 10-digit Indian mobile number (`+91` pre-filled).
  - Instant auto-fill of Name, Age, Gender, and UHID if patient has visited before.
  - If new, quick inline entry of Name, Age (years/months), and Gender.
- **FR-1.2 Payment Capture (Cash & Dynamic UPI):**
  - One-tap toggle: `Cash` vs `UPI`.
  - Dynamic UPI QR generation with the clinic's custom VPA (`upi://pay?pa=doctor@upi&am=500&pn=ClinicName&tr=TOKEN123`).
  - Instant confirmation mark: "Paid ₹500".
- **FR-1.3 Token Dispatch (WhatsApp & Thermal Slip):**
  - Automatic WhatsApp template trigger via WhatsApp Cloud API:

    > *"Namaste [Patient Name], your Token #[TokenNumber] at [Clinic Name] is confirmed. Currently serving: Token #[CurrentToken]. Live queue: https://clinik.in/q/[TokenId]"*

  - Direct print support for 2-inch and 3-inch thermal POS slip printers (ESC/POS via Web Bluetooth / USB).

- **FR-1.4 Priority Triage:**
  - Categorization: `Normal (Walk-in)`, `Appointment`, `Senior Citizen / Infant`, `Emergency`.

## Epic 2: The Doctor Cabin Mobile/Tablet Controller

- **FR-2.1 Active Queue Ribbon:**
  - Sticky header on doctor's mobile device displaying:
    - `[Current Patient Inside]` with live timer.
    - `[Next Patient in Line]`.
    - Prominent action buttons: `[ Call Next ]`, `[ Put on Hold (Lab) ]`, `[ Complete Consultation ]`.
- **FR-2.2 Patient Clinical Snapshot:**
  - Sliding drawer or single-screen card showing:
    - Age, Gender, Blood Group, Drug Allergies (in bold red).
    - Timeline of last 5 clinic visits and prescriptions.
- **FR-2.3 Quick-Tap Vitals Entry:**
  - Numeric quick-entry for: BP (`120/80`), Pulse (`bpm`), Weight (`kg`), Height (`cm`), Temp (`°F`), Blood Sugar (`mg/dL`).
  - Auto-calculated BMI and Pediatric percentile markers.
- **FR-2.4 Indian-Format Prescription Engine:**
  - Autocomplete search for Indian brand names and generics (e.g., *Augmentin 625*, *Pan-D*, *Calpol 250*, *Azithral 500*).
  - Dosage chips: `1-0-1`, `1-0-0`, `0-0-1`, `1-1-1`, `SOS`.
  - Meal timing chips: `After Food (Khane ke baad)`, `Before Food (Khali pet)`.
  - Duration chips: `3 Days`, `5 Days`, `7 Days`, `15 Days`, `1 Month`.
- **FR-2.5 One-Tap Rx Dispatch:**
  - Tap `[Finish & Share]`: Compiles clean digital PDF with clinic letterhead, doctor registration number, signature, and dispatches directly to patient's WhatsApp.

## Epic 3: The "Report Check" (Re-Entry) Sub-Queue

- **FR-3.1 Put on Diagnostic Hold:**

    - Doctor sends patient for blood test, ECG, or X-ray by tapping `[Hold for Diagnostics]`.
    - Patient status moves to `AWAITING_REPORTS`. The doctor is immediately prompted to call the next waiting patient.
- **FR-3.2 Patient Returns with Reports:**
    - Receptionist taps `[Mark Reports Arrived]`.
    - System places the patient in the prioritized `REPORT_REVIEW` slot (interleaved right after the current consultation concludes).
    - No extra consultation fee is billed.

## Epic 4: Zero-Friction Patient Live Tracker (Mobile PWA)

- **FR-4.1 Lightweight Web App:**
    - URL format: `clinik.in/q/{token_uuid}`.
    - Total payload size under 150KB for fast loading on 3G/4G networks.
- **FR-4.2 Real-Time Waiting Meter:**
    - Animated visual counter: "Tokens ahead of you: 3".
    - Dynamic Wait Time: "Estimated consultation time: 6:45 PM (~20 mins)".
    - Live status indicator: `Waiting in Hall` → `Please be near Cabin` → `Inside Doctor's Cabin`.
- **FR-4.3 "Running 15 Mins Late" Button:**
    - Patient can tap once to notify the clinic and automatically swap their slot with the next token without losing their place entirely.

## Epic 5: Waiting Room TV Display & Vernacular Audio Chime

- **FR-5.1 High-Contrast Fullscreen TV Display:**
    - Large, bold typography readable from 25 feet away.
    - Shows: "NOW CALLING: Token #14 — Rahul Kumar → Cabin 1".
    - Upcoming list: Next 4 tokens.
- **FR-5.2 Vernacular Voice Chime (Web Speech / Synthesizer):**
    - Soft harmonic chime sound followed by spoken announcement:
    - English: "*Token number 14, please proceed to Doctor's Cabin.*"

    - Hindi: "*टोकन नंबर 14, कृपया डॉक्टर के केबिन में जाएं।*"
    - Configurable regional language support (Tamil, Telugu, Marathi, Kannada, Bengali).

### Epic 6: Cash & Daily Business Reconciliation

- **FR-6.1 Shift Settlement Summary:**
    - End-of-day report generated for the doctor:
        - Total patients seen: 42 (Walk-ins: 31, Appointments: 11).
        - Total Collections: ₹21,000 (Cash: ₹9,500, UPI: ₹11,500).
        - Average consultation time: 6.2 minutes.
        - No-shows / Abandoned: 3.
- **FR-6.2 Export to WhatsApp / Excel:**
    - One-tap export to doctor's personal WhatsApp number at clinic close.

## 5. Non-Functional Requirements (NFRs)

| Category | Requirement | Target Metric |
|---|---|---|
| **Performance** | Token creation time | < 500ms API response; < 12s user interaction |
| **Real-time Sync** | Latency between Doctor tap and TV screen update | < 300ms via WebSockets / BroadcastChannel |
| **Offline Resilience** | Local browser/LAN storage survival during internet drops | Full read/write operation for up to 8 hours offline |
| **Security & Privacy** | Patient data encryption in transit & rest | TLS 1.3, AES-256 for health records, DISHA/ABDM compliant |
| **Accessibility** | Doctor mobile UI accessibility | High contrast, min 48px touch targets, single-hand thumb reach |

## 6. Success Metrics & KPIs

1. **Reception Bottleneck:** Front-desk check-in time reduced from ~120s (manual register) to under 15s.
2. **Waiting Room Congestion:** Average physical waiting room crowd reduced by 40% (patients arrive just in time based on WhatsApp ETA).
3. **Doctor Productivity:** Doctor saves ~30 seconds per patient consultation compared to traditional desktop EMRs.
4. **Patient Satisfaction:** Patient Net Promoter Score (NPS) > 75 for clinic waiting experience.